NIVXRAY

TECHNICAL DEMO

v2.0 · Feb 2026

TECHNICAL WALKTHROUGH

Inside the pipeline: archetypes, magic race, MoE panel, chain aggregator.

97 ops · 40+ wrappers · 12 threat-intel sources · one deterministic recipe.

Deep-dive · 45 min · live payload run

ARCHITECTURE

React + FastAPI + MongoDB. Everything else is optional.

Stateless API, single Mongo, pluggable LLM. Runs on 2 vCPU / 4 GB in prod.

FRONTEND	React 19 · shadcn/ui · Cytoscape (attack graph) · React-Router
API	FastAPI · Pydantic v2 · JWT auth · SSE streaming for long LLM calls
CORE	Wrapper archetypes · Magic decoder · MoE panel · Threat-model assessor
DATA	MongoDB (investigations, users, corrections, KB, feeds)
INTEGRATIONS	Emergent LLM key (Claude 4.5 Sonnet) · VirusTotal · OTX · AbuseIPDB · URLScan · Shodan

DECODING PIPELINE

Deterministic first. LLM only when confidence < 40 %.

6 gated stages. Every stage produces a machine-checkable recipe.

1. PRE-CLEAN	PS variable resolver · unicode normalisation · quote-mangling fixups
2. ARCHETYPES	40+ named wrapper handlers (Empire, Cobalt, MSF, Bash gzip pipe, Node Buffer, …)
3. MAGIC RACE	Candidate scoring across base64/hex/url/gzip/zlib/lzma/bzip2/xor/rc4/utf16
4. BOOST	Recursive re-entry on decoded output (max depth 6) — catches nested stagers
5. AGGREGATE	Merge IOCs / MITRE / LOLBAS / YARA / kill-chain across stages
6. LLM NARRATIVE	ONE call across the full aggregate → analyst-grade description + verdict

Named handlers beat generic decoders every time.

Each archetype = regex + fixed chain + pytest regression. Chain-of-archetypes to depth 4.

- **PS_MemoryStream_Gzip_IEX**
  Empire / Cobalt one-liner
- **PS_MemoryStream_Deflate_IEX**
  Deflate variant
- **PS_FromBase64String_UTF16LE**
  -EncodedCommand
- **PS_MSF_XOR_Stage2**
  Meterpreter reflective loader
- **PS_ASCII_XOR_IEX**
  (int,int,⋯) | %{[char]($_ -bxor k)}
- **PS_BINARY_SPLIT_TOINT16**
  Invoke-Obfuscation binary array
- **PS_STRING_CONCAT / -join / -f**
  Split-join / format obfuscation
- **PS_REVERSE_STRING**
  -join ('...'[-1..-N])
- **Bash_base64_gunzip_pipe**
  echo <b64> | base64 -d | gunzip | bash
- **Bash_base64_pipe_bash**
  echo <b64> | base64 -d | bash
- **Node_Buffer_from_gunzip**
  Buffer.from(<b64>,'base64') + zlib.gunzipSync
- **JS_STRING_FROMCHARCODE**
  SocGholish / Fake-Update injects
- **BATCH_VAR_SLICE**
  @set v=⋯ & %v:~x,y%
- **PS variable indirection**
  Feb-2026 · resolve $b='⋯' before match
- **Corrupt-gzip salvage**
  Feb-2026 · raw-DEFLATE fallback on bad CRC
- **Chain-of-archetypes (depth 4)**
  Stage-2 unwraps into Stage-3 automatically

MULTI-STAGE CHAIN ANALYSER

One paste. N stages. One SOC verdict.

POST /api/decode/chain aggregates per-stage findings + amplifies risk on chain length.

```
$ curl -X POST $API/api/decode/chain \
    -H "Authorization: Bearer $JWT" \
    -d '{
      "stages": [
        {"input": "powershell.exe -NoP …"},
        {"input": "certutil -urlcache -f …"},
        {"input": "vssadmin delete shadows …"}
      ]
    }'

→ stages[3] · aggregate:
    family: Destructive Wiper / Ransomware Precursor
    verdict: Malicious · 100/100 · chain-amplified
    kill_chain: [T1059.001, T1140, T1490, T1485]
    lolbas: 6 unique wiper binaries
```

- **Blank-line paste auto-splits.**
  splitCommandLines() + newline heuristics.
- **Per-stage RE-RUN.**
  Edit a stage, replay only the tail — full re-aggregate coming Q2.
- **AI narrative on WHOLE chain.**
  ONE LLM call · SSE-streamed · Cloudflare-524 immune.
- **Export .MD / .JSON / STIX 2.1 (P1).**
  Every verdict shippable to a ticket in one click.

MOE ANALYST PANEL

Three critics + a synthesizer. Disagreement is signal.

Runs on the FULL aggregate. Emits consensus + disagreements as first-class fields.

MALWARE ANALYST	Family attribution · packer detection · capability tags
RED-TEAM CRITIC	Evasion techniques · living-off-the-land opportunities
DEFENSIVE CRITIC	Detection gaps · response actions · containment steps
SYNTHESIZER	Consensus + explicit disagreements + confidence dial

Anti-hallucination: synthesizer refuses claims not backed by aggregate evidence.

Analyst corrections feed the deterministic layer.

4-verdict picker + versioned overrides + admin approval queue.

INCORRECT

Deterministic override — future analyses drop the wrong finding.

PARTIAL

Steer the LLM without deleting the finding.

CORRECT

Positive reinforcement — no override, tracked for accuracy.

SUGGEST

Advisory improvement — LLM-inject only, never blocking.

Admin dashboard at /admin/corrections shows: 10 metric buckets, by-surface heatmap, top reused, FP/FN signal, reviewer stats, avg approval time, 7-day trend.

THREAT-INTEL ENRICHMENT

Every extracted IOC is enriched — no manual pivoting.

Parallel calls to 8+ sources, deduped, cached, defanged for safe embed in reports.

- **VirusTotal**
  hash · URL · IP · domain reputation + AV verdicts
- **AlienVault OTX**
  pulses · adversary campaigns · TTPs
- **AbuseIPDB**
  IP abuse score · report count · category
- **URLScan.io**
  screenshots · DOM · redirect chain
- **Hybrid Analysis**
  sandbox verdict · related samples
- **Shodan**
  port · service · certificate fingerprint
- **GreyNoise**
  internet-noise classification (bg/bening/malicious)
- **IPinfo / abuse.ch**
  ASN / geo · Feodo / URLhaus / SSLBL feeds

PUBLIC API

One JWT. Nine core endpoints.

Full OpenAPI spec at /docs. Every response includes recipe + trace for replay.

```
POST  /api/auth/login                → JWT + role
POST  /api/auth/change-password      → force-rotate seeded creds

POST  /api/decode/smart              → hybrid archetype+magic race
POST  /api/decode/chain              → N stages · aggregate verdict
POST  /api/decode/chain/narrative    → ONE LLM narrative on full chain
POST  /api/decode/chain/export       → .md / .json / .stix

POST  /api/threat-model/analyze      → Mermaid → threat findings
GET   /api/enrich/ioc                → VT / OTX / Abuse / URLScan / ···

GET   /api/corrections/analytics     → admin · 10 metric buckets
POST  /api/corrections               → analyst refine (4-verdict)
GET   /api/investigations/{id}       → owner-scoped replay
```

QUALITY SIGNAL

Every archetype pinned by a pytest regression.

Deterministic layer is auditable — you can prove why a verdict fired.

115+	40+	97	0
PYTEST CASES	WRAPPER ARCHETYPES	DECODER OPS	KNOWN FALSE-POSITIVE FAMILIES

- **Regression matrix.**
  Every real-world captured payload becomes a pinned test — never breaks silently.
- **Corpus fuzzing.**
  training/corpus/generator_v2.py — 900+ synthetic obfuscation shapes.
- **SEC audit passed.**
  SEC-001/002/003 remediated: JWT secret rotated, admin seed idempotent, owner-scoped investigations.

LIVE DEMO · 5 MIN

Watch the pipeline light up in real time.

Everything below is scripted so it always works — but pausable at every step.

- **1. Paste PS gzip loader**
  Empire-style one-liner with variable indirection ($b='H4sI…')
- **2. Hit NIVXRAY DECODE**
  Archetype fires → stage 0 confidence 100 in < 1 s
- **3. Reveal chain-mode auto-detect**
  Multi-line paste auto-splits into 3 stages
- **4. Aggregate SOC verdict**
  Family: Destructive Wiper · Malicious 100/100 · MITRE mapped
- **5. AI NARRATIVE (whole chain)**
  ONE LLM call → analyst-grade description streamed via SSE
- **6. Export .MD**

QUESTIONS · CODE · ARCHITECTURE

Let's open the code.

We'll drop into /app/backend/wrapper_archetypes.py or wherever your team wants to go.

engineering@nivxforge.com

github.com/nivxforge/nivxray · discord.gg/nivxforge